Decomposed Weight Updates

Varga, Iyer et al. · 2025

Fictional sample paper generated for the ClaimLens demo.

The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report.

Ablations in the appendix vary one factor at a time. Results for individual tasks appear in the appendix. We release the evaluation scripts so that every number can be reproduced. We discuss the limitations of this protocol in the final section.

We evaluate the final checkpoint rather than the best one seen during training. We report wall-clock time on a single node of eight accelerators. We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps.

This setting matches the one most often used in practice. Unless noted otherwise, hyperparameters are tuned on a held-out validation split. Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models.

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We follow the standard training recipe and report the mean over three seeds. Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

Variance across seeds is small compared with the differences we report. Prior work reached similar conclusions on smaller models. All experiments use a 7B base model. Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup.

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We discuss the limitations of this protocol in the final section. We keep the tokenizer and context length fixed across every configuration. Table 1 reports accuracy on a commonsense reasoning suite. We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant.

Table 1

Each configuration is trained for the same number of optimizer steps. Training uses mixed precision with gradient clipping at a norm of one. Accuracy is averaged over the eight commonsense tasks. Training uses mixed precision with gradient clipping at a norm of one. Ablations in the appendix vary one factor at a time.

Prior work reached similar conclusions on smaller models. All runs share the same learning-rate schedule with a short linear warmup. With magnitude decomposition, the adapter closes the remaining gap on commonsense reasoning. All runs share the same learning-rate schedule with a short linear warmup. We evaluate the final checkpoint rather than the best one seen during training.

We keep the tokenizer and context length fixed across every configuration. The comparison holds the data and the evaluation harness constant. Decomposition improves average accuracy by 1.1 points over LoRA. The comparison holds the data and the evaluation harness constant. This setting matches the one most often used in practice.

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